

Total No. of Page(s): 02..

2nd SEMESTER

Roll No.....

B.Tech. (ECE/EVDT/ICE)

MID SEM. EXAMINATION, Feb.-2025

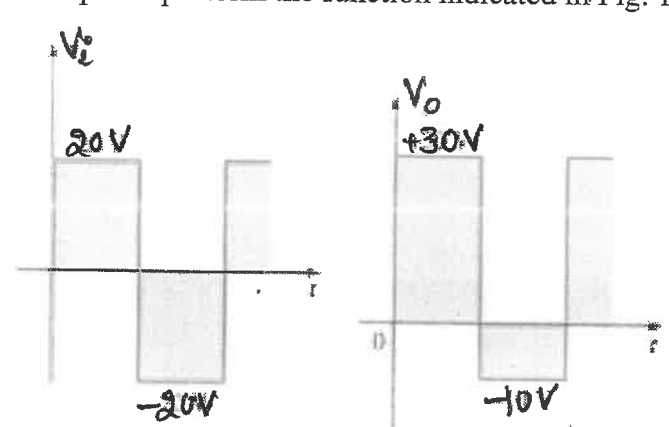
Course Code: ECECC202/ VTECC202/ICECC202/EAEP202

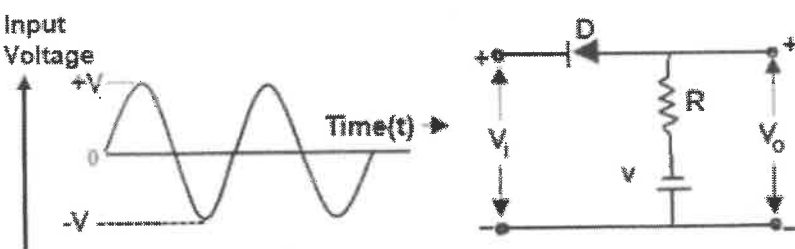
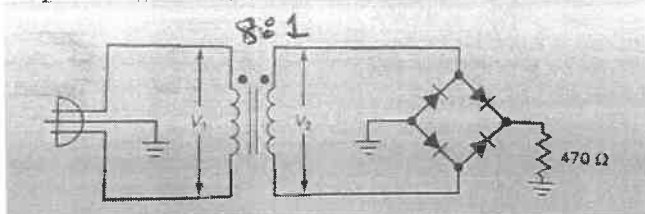
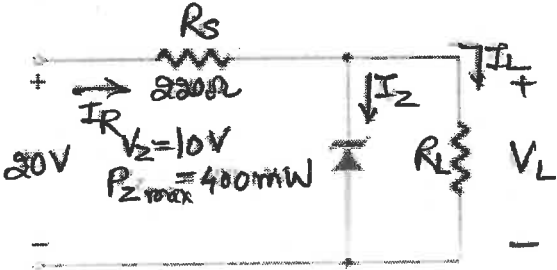
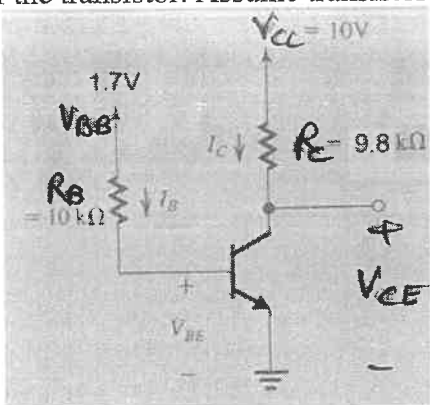
Course Title: Electronic Devices and Circuits

Time: 01:30 Hrs.

Max. Marks: 20

Note: Attempt all questions. All questions carry equal marks. Missing data/information if any will be suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO
1(a)	Derive the relation of barrier potential for a step-graded p-n junction.	2	CO1
1(b)	Determine the resistivity of germanium (a) in intrinsic condition at 300 K (b) with donor impurity of 1 in 10^7 (c) with acceptor impurity of 1 in 10^8 (d) with both the above impurities simultaneously. Given that for germanium at room temperature $n_i = 2.5 \times 10^{13}/\text{cm}^3$, $\mu_n = 3800 \text{cm}^2/\text{Vs}$, $\mu_p = 1800 \text{cm}^2/\text{Vs}$, and a number of Germanium atoms/ $\text{cm}^3 = 4.4 \times 10^{22}$.	2	CO1
2(a)	Define the diffusion and transition capacitances of a p-n junction and show their variation with changing bias voltage.	2	CO2
2(b)	Design a clamper to perform the function indicated in Fig. 1  (a) Input volatge (b) Output voltage Fig. 1 *Here, v_i = input volatgeg and v_o = output voltage	2	CO2
3(a)	For the clipper circuit shown in Fig. 2, draw the output waveform obtained when input voltage (V) = 10V and battery in series with R (v) = 2V.	2	CO2

	 Fig. 2		
3(b)	For the Bridge Rectifier shown in Fig. 3, calculate: (a) Peak output voltage if diodes are ideal, (b) Output DC voltage, (c) Average output voltage and (d) PIV across diodes, provided line voltage = 125Vrms.  Fig. 3	2	CO2
4(a)	Determine the minimum value of R_L to ensure that the Zener diode is in the "on" state.  Fig. 4	2	CO2
4(b)	Explain the concepts of Zener breakdown and Avalanche breakdown. Describe the underlying phenomena and the conditions under which each type of breakdown occurs.	2	CO2
5(a)	Draw the output characteristics of an NPN transistor in the common base (CB) configuration. Clearly label the regions of operation on the characteristic curve.	2	CO3
5(b)	For the circuit shown in Fig. 5, determine the operating region and operating point of the transistor. Assume transistor $\beta = 50$ (at least).  Fig. 5	2	CO3